

Robotics and Digital Book Project

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Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 2
- **Grade Hub:** Notes [../grade-2-Notes.md]
- **Schedule Reference:** Spring2026_TeachingSchedule_Derek.md [../school/Spring2026_TeachingSchedule_Ignite/grade-2/Spring2026_TeachingSchedule_Derek.md]
- **Workbook Sheet:** 2nd in Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx
- **Lesson Focus:** Students combine simple robotics routines with digital storytelling or explanatory book creation, linking movement logic to written or multimedia communication.
- **CS Alignment Strength:** Strong CS lesson with literacy integration
- **LaTeX Source:** robotics-and-digital-book-project.tex [../robotics-and-digital-book-project.tex]
- **Bib File:** robotics-and-digital-book-project.bib [../robotics-and-digital-book-project.bib]
- **Artifacts Folder:** artifacts/ [../artifacts/]

Lesson Identity

- **Likely Class Blocks:** Day 1 – 10:45-11:35 – Callon, Day 1 – 2:15-3:05 – Hennen, Day 4 – 10:45-11:35 – Barthelman
- **Main Lesson Family:** Robotics and Digital Book Project
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- sequencing robot moves or routes
- connecting a digital page or book entry to a robotics task
- describing what a robot does and why
- using technology to document or explain a process

CS Standards Alignment

- **1A-AP-10:** Develop programs with sequences and simple loops to express ideas or address a problem. Why it fits here: Students create or test simple robot movement sequences.
- **1A-AP-14:** Debug errors in an algorithm or program that includes sequences and simple loops. Why it fits here: Students revise the route or command order when the robot does not do what they expected.
- **1A-IC-17:** Work respectfully and responsibly with others when using shared technology. Why it fits here: The lesson depends on pair work around devices, robots, and shared book materials.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 2 IGNITE workbook through robotics plus digital composition.
- Supports New York-facing expectations for digital communication by requiring students to explain or document what the robot did.
- The lesson counts as CS because students create and revise step-by-step computational behavior, then represent that process in a digital artifact.

Lesson Content and Concepts

- sequence
- debugging
- robot behavior
- digital artifact
- explanation

Evidence / Proof of Instruction

- **Transcript-derived note:** 260225 Grade 2 Callon Sphero Book [[../transcript-derived-lessons/260225-grade-2-callon-sphero-book.md](#)] The main note links the lesson directly to Sphero Indi and Book Creator as paired parts of the same class experience.

- **Transcript-derived note:** 260227 Grade 2 Callon Creator Onboarding [../transcript-derived-lessons/260227-grade-2-callon-creator-onboarding.md] Follow-up onboarding and library verification notes show that the digital book workflow was actively taught, not merely assigned.
- **Transcript-derived note:** 260225 Grade 2 Callon Book Circle [../transcript-derived-lessons/260225-grade-2-callon-book-circle.md] The class review indicates that explanation, reflection, and classroom discussion were part of how students processed the task.
- **Likely student proof:** Robot paths, Book Creator pages, screenshots, or student explanation pages all count as lesson evidence.

Assessment Evidence

- Student can make a robot follow a basic route or command set.
- Student can explain the route or task in a digital page or oral explanation.
- Student can revise either the robot behavior or the digital description when something is missing or unclear.

UDL and Inclusion Supports

Multiple Representation

- Model both the physical route and the digital page side by side.
- Use arrows, icons, and color-coded path markers.
- Keep page expectations low-text and high-visual when needed.

Multiple Action / Expression

- Allow students to demonstrate understanding through route setup, oral recording, drawing, or typed text.
- Use partner roles such as driver, builder, recorder, and checker.
- Offer sentence stems for explaining what the robot should do.

Multiple Engagement

- Combine movement and storytelling so students can enter through either robotics or communication.
- Celebrate each successful route or finished page as a milestone.
- Use authentic audience language: this page helps someone else understand the robot.

How We Are Already Making It Inclusive

- The transcript trail shows the lesson was already integrating multiple modes, not relying on one narrow output.
- Teacher onboarding support reduced the access barrier for the book-creation platform.
- The robotics component created a concrete anchor for students who need physical demonstration before abstract explanation.

How to Improve Inclusion Next Time

- Add a one-page route planner students complete before touching the robot.
- Prepare a low-text book template with image-first boxes.
- Collect one exemplar page and one exemplar route photo in **artifacts/**.
- Build in a final comparison step: what the robot did versus what the page says it did.

Materials and Setup

- Robot set or Sphero Indi
- Student devices
- Book Creator or similar digital book platform
- Route cards or floor path markers

Teacher Moves / Script / Routines

- Model the route physically before opening the digital explanation task.
- Name one debugging move students must try before asking for help.
- Pause to show a strong example of a page that clearly explains the robot action.
- Close with a short share-out of what changed after debugging.

Common Errors and Debugging

- Students may focus on the robot only and skip the explanation side.
- Students may make the digital page decorative but not informative.
- Pair work can drift if roles are not explicit.

Extensions and Simplifications

- Extension: ask students to create a second page about how they fixed a mistake.
- Simplification: provide a preset route and ask students only to explain it.
- Extension: add a “predict before test” step for each route.

Related Local Materials

- No direct local lesson packet linked yet.

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References